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Quantum Waves in Motion: Numerical Modeling of the Time-Dependent Schrödinger Equation

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1 Introduction

The Schrödinger Equation is one of the foundations of quantum mechanics; it helps us understand how a quantum system evolves. This document provides a step-by-step guide to a numerical solution of the time-dependent Schrödinger Equation (TDSE) using a finite-domain time-difference method. We aim to demystify the approach by dividing the problem into small pieces and providing visualisations that give insight into wave packet dispersion, reflection and tunnelling. In particular, we adopt techniques similar to those described by Robertson (2011) and Visscher (1991). Similar approaches have been discussed by Feit et al. (1982) and Tannor (2007). Press et al. (1992) provides a straightforward introduction to finite-difference schemes and boundary condition handling.

2 The Time-Dependent Schrödinger Equation

The TDSE describes the time evolution of the wavefunction $\Psi(x,t)$ of a quantum particle:

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = H(x)\Psi(x,t), \tag{1}$$

where:

- $\hbar$ is the reduced Planck constant,
- i is the imaginary unit, and
- $H(x)$ is the Hamiltonian operator.

2.1 The Hamiltonian Operator

The Hamiltonian, representing the total energy, comprises two parts:

- **Kinetic Energy:** Given by

  $$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}, \tag{2}$$

  where m is the mass of the particle.
- **Potential Energy:** Represented by $V(x)$, which characterises the external potential landscape.

Thus, the Hamiltonian is written as:

$$H(x) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x). \tag{3}$$

3 Numerical Approach: Discretisation of Time and Space

Analytical solutions of the TDSE are rare, so we use a numerical method that discretizes both time and space.

3.1 Time Discretization

Time is divided into small increments Δt. The time derivative is approximated as:

$$\frac{\partial \Psi(x,t)}{\partial t} \approx \frac{\Psi(x,t+\Delta t) - \Psi(x,t)}{\Delta t}. \tag{4}$$

3.2 Space Discretisation

Similarly, space is divided into segments of length Δx, and the second derivative is approximated by:

$$\frac{\partial^2 \Psi(x,t)}{\partial x^2} \approx \frac{\Psi(x+\Delta x,t) - 2\Psi(x,t) + \Psi(x-\Delta x,t)}{(\Delta x)^2}. \tag{5}$$

4 Update Rule for the Wavefunction

By substituting the finite-difference approximations (4) and (5) into Equation (1), we derive an iterative update rule:

$$\begin{aligned}\Psi(x,t+\Delta t) &= \Psi(x,t) - \Delta t\,[\text{spatial difference terms and } V(x)] \\ &= \Psi(x,t) - \Delta t\,\frac{\hbar}{2i\,m\,(\Delta x)^2}\,[\Psi(x+\Delta x,t) - 2\,\Psi(x,t) + \Psi(x-\Delta x,t)] \\ &\quad + \Delta t\,\frac{i}{\hbar}\,V(x)\,\Psi(x,t).\end{aligned} \tag{6}$$

For clarity, we define constants

$$c_1 = \frac{\Delta t}{2\,m\,(\Delta x)^2} \qquad \text{and} \qquad c_2 = \frac{\Delta t}{\hbar}. \tag{7}$$

so that when we separate the real and imaginary parts, the update equations become:

$$\Phi_R(x,t+\Delta t) = \Phi_R(x,t) - c_1\,[\Phi_I(x+\Delta x,t) - 2\Phi_I(x,t) + \Phi_I(x-\Delta x,t)] + c_2\,V(x)\,\Phi_I(x,t), \tag{8}$$

$$\Phi_I(x,t+\Delta t) = \Phi_I(x,t) + c_1\,[\Phi_R(x+\Delta x,t) - 2\Phi_R(x,t) + \Phi_R(x-\Delta x,t)] - c_2\,V(x)\,\Phi_R(x,t). \tag{9}$$

These equations are the foundation for our numerical propagation.

5 Initial State: The Gaussian Wave Packet

To simulate a localised quantum particle, we choose a Gaussian wave packet:

$$\Psi(x,0) = \frac{1}{\sqrt{\sigma\sqrt{\pi}}} \exp\left(-\frac{(x-x_0)^2}{2\sigma^2}\right). \tag{10}$$

The real and imaginary parts are obtained by modulating this envelope with cosine and sine functions:

$$\Phi_R(x,0) = \frac{1}{\sqrt{A}} \exp\left(-\frac{(x-x_0)^2}{2\sigma^2}\right)\cos\Big[k(x-x_0)\Big], \tag{11}$$

$$\Phi_I(x,0) = \frac{1}{\sqrt{A}} \exp\left(-\frac{(x-x_0)^2}{2\sigma^2}\right)\sin\Big[k(x-x_0)\Big], \tag{12}$$

where A is determined via the Born rule:

$$\int_{-\infty}^{\infty} |\Psi(x,0)|^2 \, dx = 1\,. \tag{13}$$

6 Exploring Different Potentials

One of the appealing aspects of this simulation is how easy it is to change the potential $V(x)$. Common choices include:

- **Finite Square Well:** The potential is lower in a finite region.
- **Harmonic (Quadratic) Well:** A parabolic potential ideal for modeling oscillators.
- **Finite Step Well:** A sudden jump in potential.

We study various physical scenarios by altering $V(x)$ in Equation (3).

7 Simulation and Visualisation

The numerical update (9) is applied iteratively to evolve the wave function. At each time step, we compute the probability density:

$$|\Psi(x,t)|^2 = \Phi_R(x,t)^2 + \Phi_I(x,t)^2\,. \tag{14}$$

This data then produces plots and animations, revealing how the wave packet spreads, reflects, and tunnels under different potentials.

8 Conclusion

To conclude, my project allows us to substantiate the abstract of the TDSE itself in a more practical hands-on simulation. One can have a clear picture of quantum dynamics by taking time and space to discrete values, separating the wave function into its real and imaginary parts, and evolving it step by step. This flexibility in exploring diverse scenarios not only deepens our theoretical understanding but also provides an intuitive grasp of complex phenomena such as interference, reflection, and tunneling.

References

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